

SCIENCE — Q.P. Code 31/1/3 (Set ~3)

Section A — Questions 1 to 8 (Multiple Choice, 1 mark each)

1. The metals obtained from their molten chlorides by the process of electrolytic reduction are :

- (A) Gold and silver
(B) Calcium and magnesium
(C) Aluminium and silver
(D) Sodium and iron

2. The formation of magnesium oxide is correctly shown in option :

- (A) $\text{Mg} \cdot + \text{O} \cdot \cdot \rightarrow \text{Mg}^{2+}[\text{O}^{2-}]$
(B) $\text{Mg} \cdot + \text{O} \cdot \cdot \rightarrow \text{Mg}^+[\text{O}^-]$
(C) $2 \text{Mg} \cdot + \text{O} \cdot \cdot \rightarrow \text{Mg}^{2+}[\text{O}^-]_2$
(D) $2 \text{Mg} + \text{O} \cdot \cdot \rightarrow [\text{Mg}^{2+}]_2[\text{O}^{2-}]$

(Electron-dot structures simplified from the original diagram for text reproduction.)

3. Reaction between two elements A and B, forms a compound C. A loses electrons and B gains electrons. Which one of the following properties will **not** be shown by compound C ?

- (A) It has high melting point.
(B) It is highly soluble in water.
(C) It has weak electrostatic forces of attraction between its oppositely charged ions.
(D) It conducts electricity in its molten state or aqueous solution.

4. Consider the following reactions :

- (i) Dilute hydrochloric acid reacts with sodium hydroxide.
(ii) Magnesium oxide reacts with dilute hydrochloric acid.
(iii) Carbon dioxide reacts with sodium hydroxide.

It is found that in each case :

- (A) Salt and water is formed.
(B) Neutral salts are formed.
(C) Hydrogen gas is formed.
(D) Acidic salts are formed.

5. Tooth enamel is made up of calcium hydroxyapatite (a crystalline form of calcium phosphate). This chemical starts corroding in the mouth when the pH is :

- (A) 7
(B) 5
(C) 10
(D) 14

6. The products formed when Aluminium and Magnesium are burnt in the presence of air respectively are :

- (A) Al_3O_4 and MgO_2
(B) Al_2O_3 and MgO
(C) Al_3O_4 and MgO
(D) Al_2O_3 and MgO_2

7. Electrolysis of water is a decomposition reaction. The mass ratio ($M_{\text{H}} : M_{\text{O}}$) of hydrogen and oxygen gases liberated at the electrodes during electrolysis of water is :

- (A) 8 : 1
(B) 2 : 1
(C) 1 : 2
(D) 1 : 8

8. The breakdown of glucose has taken the following pathway :

Glucose $\rightarrow^{(a)}$ Pyruvate + Energy $\rightarrow^{(b)}$ Lactic acid + Energy

The sites 'a' and 'b' respectively are :

- (A) Mitochondria and Oxygen deficient muscle cells
(B) Cytoplasm and Oxygen rich muscle cells
(C) Cytoplasm and Yeast cells
(D) Cytoplasm and Oxygen deficient muscle cells